

Royal Trees

Given N , A , B and D , find number of trees of size N in which $\text{dist}(A, B) = D$.

Subtask 2

$D = 1$.

$$\# \text{trees} = (N - 1)!$$

Since B only has one choice

$$\# \text{trees} = \frac{(N - 1)!}{B - 1}$$

Subtask 3

$A = 1, D = 2$.

Let Q denote the node between A and B .

$$\begin{aligned} \# \text{trees} &= \sum_{Q=2}^{B-1} \frac{(N - 1)!}{(Q - 1)(B - 1)} \\ &= \frac{(N - 1)!}{B - 1} \sum_{Q=2}^{B-1} \frac{1}{Q - 1} \end{aligned}$$

Subtask 4

$D = 2$. In all subtasks, $\text{dist}(A, r) + \text{dist}(B, r) = D$ for a given node r . Cases:

- $\text{dist}(A, r) = \text{dist}(B, r) = 1$:

$$\begin{aligned} \# \text{trees}_1 &= \sum_{r=1}^{A-1} \frac{(N - 1)!}{(A - 1)(B - 1)} \\ &= \frac{(N - 1)!}{(A - 1)(B - 1)} (A - 1) \\ &= \frac{(N - 1)!}{B - 1} \text{ unless } A = 1 \end{aligned}$$

- $r = A$:

Let Q denote the node between A and B .

$$\begin{aligned} \# \text{trees}_2 &= \sum_{Q=A+1}^{B-1} \frac{(N - 1)!}{(Q - 1)(B - 1)} \\ &= \frac{(N - 1)!}{B - 1} \sum_{Q=A+1}^{B-1} \frac{1}{Q - 1} \end{aligned}$$

So,

$$\begin{aligned} \# \text{trees} &= \frac{(N - 1)!}{B - 1} + \left(\frac{(N - 1)!}{B - 1} \sum_{Q=A-1}^{B-1} \frac{1}{Q - 1} \right) \\ &= \frac{(N - 1)!}{B - 1} \left[\{1 \text{ if } A \neq 1\} + \sum_{Q=A+1}^{B-1} \frac{1}{Q - 1} \right] \end{aligned}$$

Subtask 5

$A = 1$.

Let the nodes $Q_1, Q_2, \dots, Q_{k-1}, Q_k$ denote the $k = D - 1$ nodes between A and B .

$$\# \text{trees} = \frac{(N-1)!}{B-1} \sum_{Q_1=2}^{B-k} \left(\frac{1}{Q_1-1} \sum_{Q_2=Q_1+1}^{B-k+1} \left(\frac{1}{Q_2-1} \sum_{Q_3=Q_2+1}^{B-k+2} \left(\frac{1}{Q_3-1} \dots \right) \right) \right)$$

Let

$$\begin{aligned} f(i, j) &= \sum_{Q=j+1}^{B-k+i} \frac{1}{Q-1} f(i+1, Q) \\ f(k-1, j) &= \sum_{Q=j+1}^{B-1} \frac{1}{Q-1} \\ f(i, j) &= f(i, j+1) + \frac{1}{j} f(i+1, j+1) \\ f(i, j) &= 0 \text{ if } j > B - D + i \end{aligned}$$

Therefore $\# \text{trees} = \frac{(N-1)!}{B-1} f(0, 1)$.

Full

Let r be a node such that $A, B \in \text{Desc}(r)$ and $\text{dist}(A, r) + \text{dist}(B, r) = D$.

Define $c(i)$ as:

$$c(i) = \begin{cases} 2 & \text{if } i < A \\ 1 & \text{if } A < i < B \\ 0 & \text{if } i = A \text{ or } i \geq B \end{cases}$$

Define $g(i, j)$ as where $k = D - 2$:

$$\begin{aligned} g(i, j) &= \sum_{Q=j+1}^{B-k+i} \frac{c(Q)}{Q-1} g(i+1, Q) \\ g(k, j) &= 1 \\ g(i, B-1) &= 0 \text{ if } i < k \\ g(i, j) &= g(i, j+1) + \frac{c(j+1)}{j} g(i+1, j+1) \end{aligned}$$

$$\# \text{trees} = \frac{(N-1)!}{B-1} f(0, 1) \quad [A = 1]$$

$$\# \text{trees} = \frac{(N-1)!}{(A-1)(B-1)} \sum_{r=1}^{A-1} g(0, r) + \frac{(N-1)!}{B-1} f_{D-1}(0, A) \quad [A > 1]$$